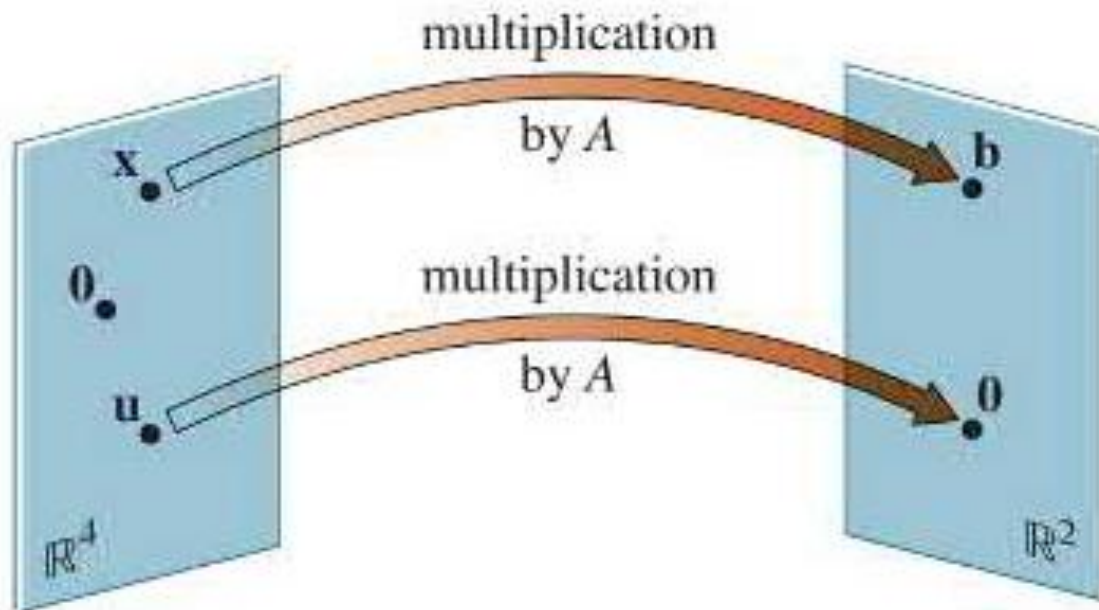


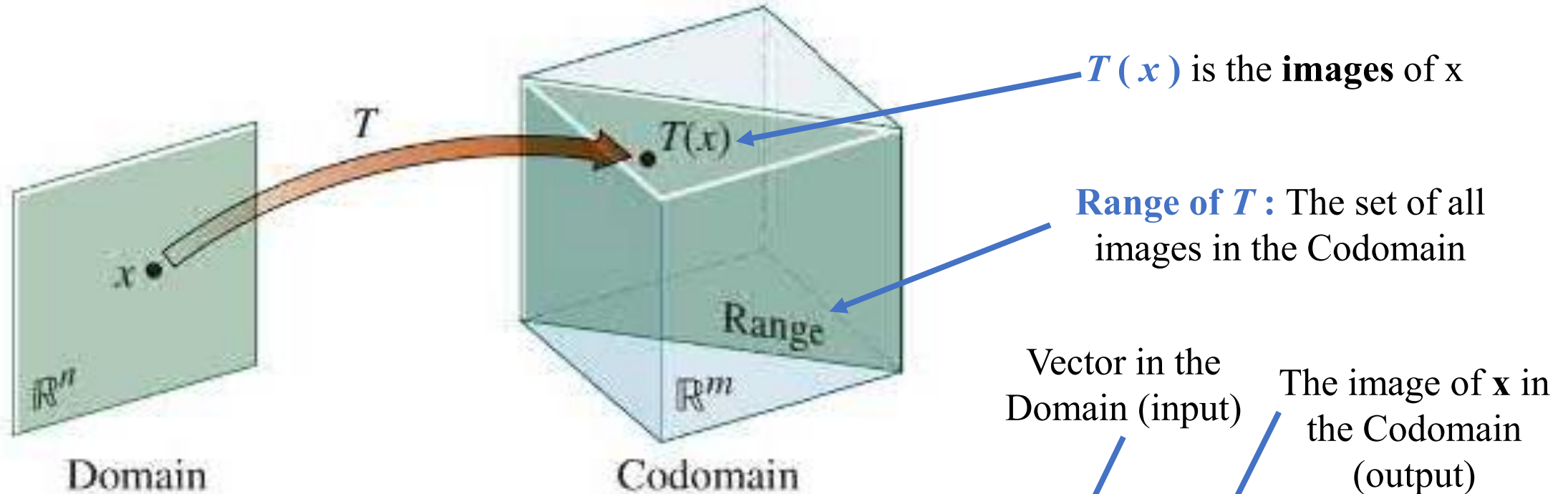
Introduction to Linear Transformation

$$\begin{array}{c} \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ A \quad \quad \quad x \quad \quad \quad b \quad \quad \quad A \quad \quad \quad u \quad \quad \quad 0 \end{array}$$



Transformation is also known as **function** or **mapping**

Introduction to Linear Transformation



This transformation is matrix-vector multiplication

$$A\mathbf{x} = \mathbf{b}$$

The standard matrix of linear transformation $m \times n$

Codomain

Domain

EXAMPLE 1 Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .

EXAMPLE 1 Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .

Compute

$$T(\mathbf{u}) = A\mathbf{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.

Solve $T(\mathbf{x}) = \mathbf{b}$ for $\mathbf{x}$. That is, solve $A\mathbf{x} = \mathbf{b}$, or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

check

b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.

Solve $T(\mathbf{x}) = \mathbf{b}$ for $\mathbf{x}$. That is, solve $A\mathbf{x} = \mathbf{b}$, or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{Solution} \quad \begin{cases} x_1 = 1.5 \\ x_2 = -.5 \end{cases} \quad \mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$$

check

$$A\mathbf{x} = \mathbf{b} \quad \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} = \mathbf{b}$$

Therefore, the vector whose image is $\mathbf{b}$ is the vector $\mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$

c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?

$$A\mathbf{x} = \mathbf{b}$$

c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?

We need to find whether $A\mathbf{x} = \mathbf{b}$ has a unique solution

We have already done that in part (b) above

either from the echelon form:

no false condition and no free variable.

Therefore, it has a unique solution.

$$\begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix}$$

OR from the final solution

$$\mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}$$

So, there is exactly one $\mathbf{x}$ whose image is $\mathbf{b}$

d. Determine if $\mathbf{c}$ is in the range of the transformation T .

$$\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

d. Determine if $\mathbf{c}$ is in the range of the transformation T .

$$\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

asking if the system $A\mathbf{x} = \mathbf{c}$ is consistent.

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

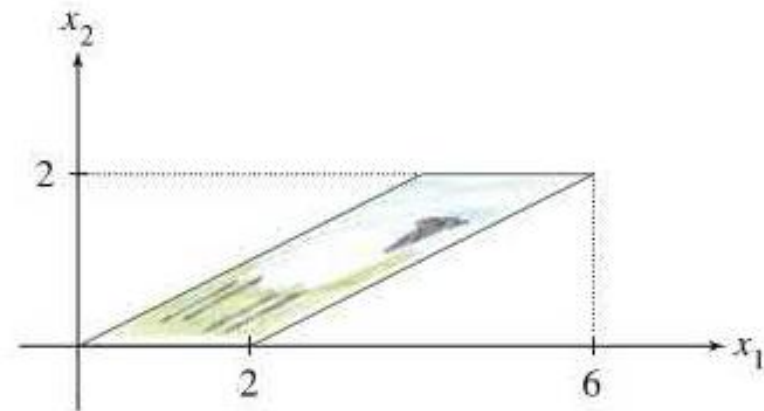
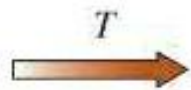
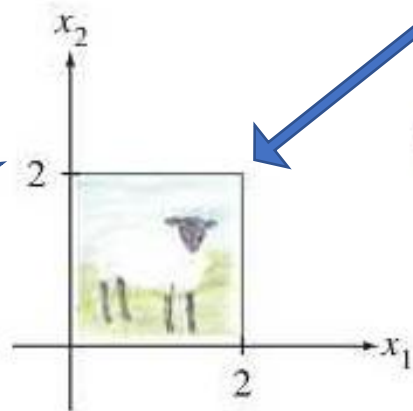
The third equation, $0 = -35$, shows that the system is inconsistent. So $\mathbf{c}$ is *not* in the range of T . 

Example: shear transformation

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(\mathbf{x}) = A\mathbf{x} \quad A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$T(\mathbf{u}) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$



Linear Transformations

A transformation (or mapping) T is **linear** if

- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
- (ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .

If T is a linear transformation, then

$$T(\mathbf{0}) = \mathbf{0}$$

and

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

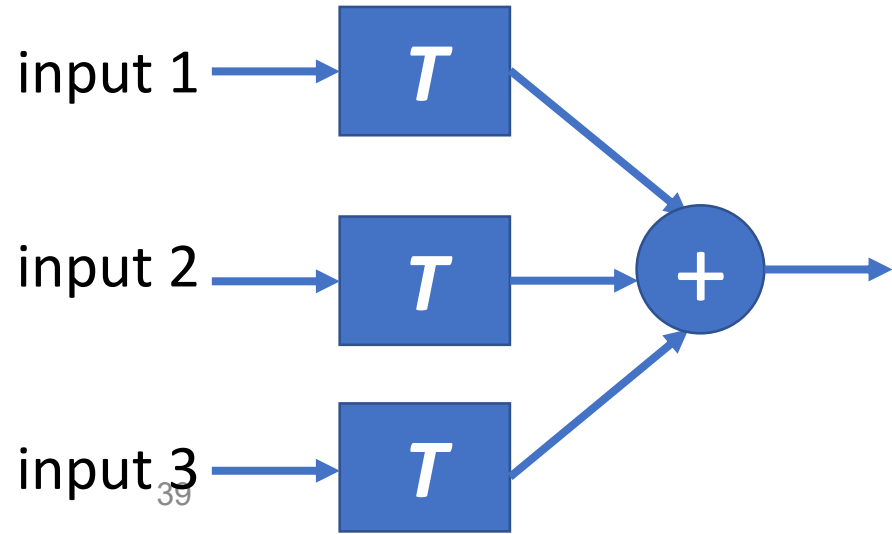
for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d .

Superposition

- Transformation of a linear combination of inputs is equal to the linear transformation of the of individual inputs

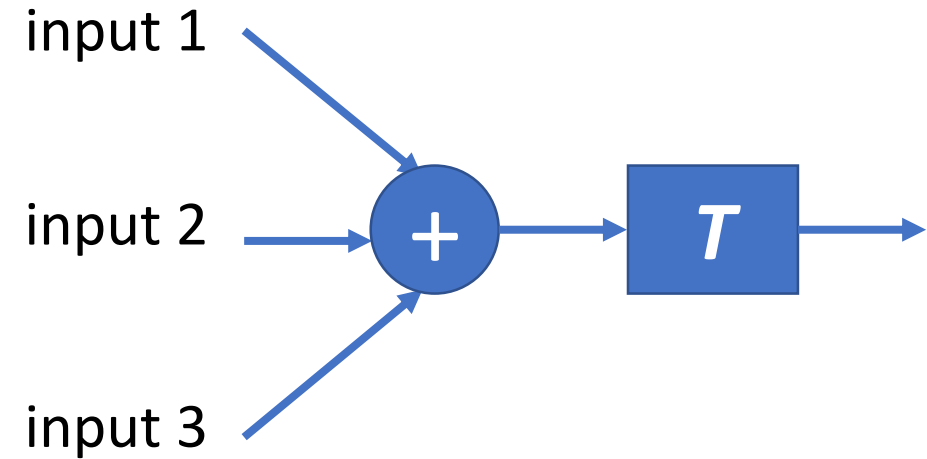
$$T(c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1) + \cdots + c_pT(\mathbf{v}_p)$$

Why Linear Transformation is useful?



```
T1 = T(input_1)
T2 = T(input_2)
T3 = T(input_3)
Output = T1 + T2 + T3
```

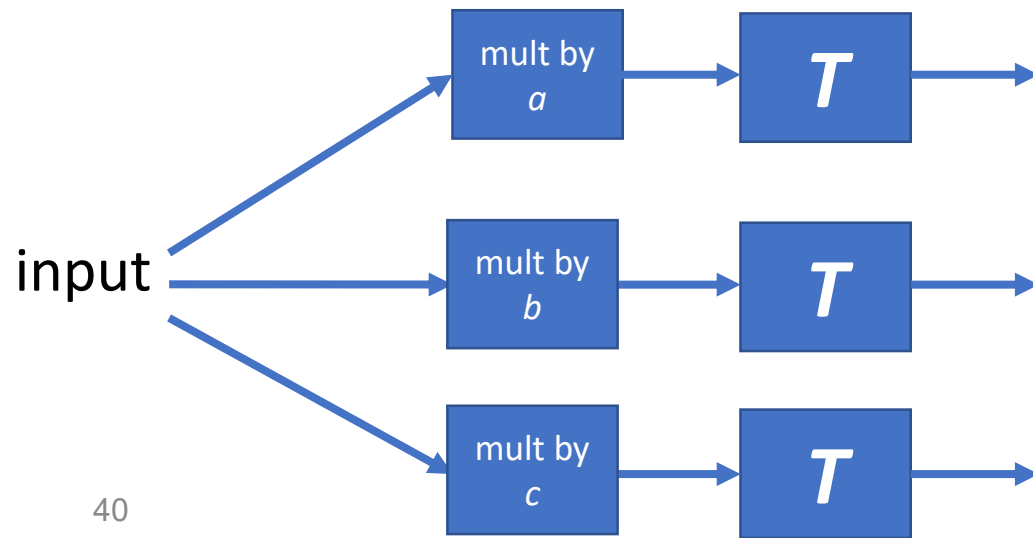
Three function calls



```
Input = input_1 + input_2 + input_3
T = T(input)
```

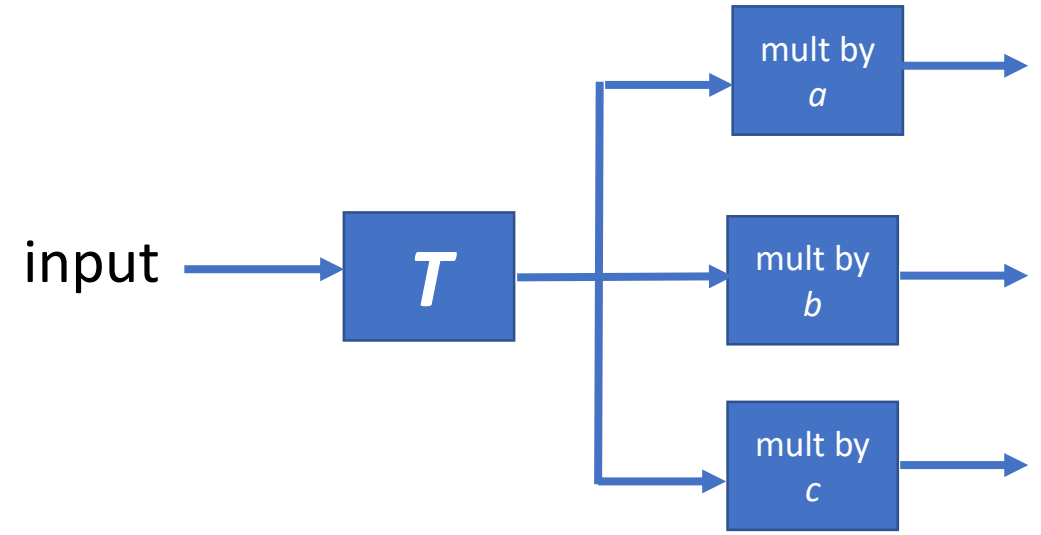
Only one function call

Why Linear Transformation is useful?



```
input_1 = a * input
input_2 = b * input
input_3 = c * input
T1 = T(input_1)
T2 = T(input_2)
T3 = T(input_3)
```

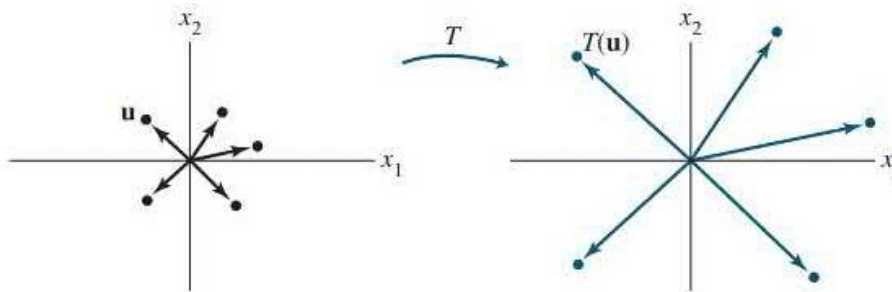
Three function calls



```
Transform = T(input)
output_1 = a * input
output_2 = b * input
output_3 = c * input
```

Only one function call

EXAMPLE 4 Given a scalar r , define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$. T is called a **contraction** when $0 \leq r \leq 1$ and a **dilation** when $r > 1$. Let $r = 3$, and show that T is a linear transformation.



We need to check the condition : $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$

$$\begin{aligned} T(c\mathbf{u} + d\mathbf{v}) &= 3(c\mathbf{u} + d\mathbf{v}) \\ &= \boxed{3c\mathbf{u} + 3d\mathbf{v}} \\ &= c(3\mathbf{u}) + d(3\mathbf{v}) \\ &= cT(\mathbf{u}) + dT(\mathbf{v}) \end{aligned}$$

Definition of T

Vector arithmetic

$$\begin{aligned} \text{scalar } c \overbrace{T(\mathbf{u})}^{T(\mathbf{x})} &= \text{scalar } c \overbrace{T(\mathbf{x})}^{T(\mathbf{x})} \\ &= \boxed{3c\mathbf{u}} \\ \text{scalar } d \overbrace{T(\mathbf{v})}^{T(\mathbf{x})} &= \text{scalar } d \overbrace{T(\mathbf{x})}^{T(\mathbf{x})} \\ &= \boxed{3d\mathbf{v}} \end{aligned}$$

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

Thus T is a linear transformation

Another solution:

This transformation is the **scalar-vector multiplication**, which has the following two properties (see this slide):

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$c(d\mathbf{v}) = (cd)\mathbf{v}$$

Algebraic Properties of $\mathbb{R}^n$
For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d :

(i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$	(v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
(ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$	(vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
(iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$	(vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$
(iv) $\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$, where $-\mathbf{u}$ denotes $(-1)\mathbf{u}$	(viii) $1\mathbf{u} = \mathbf{u}$

vector such as $\mathbf{u} + (-1)\mathbf{v}$ is often written as $\mathbf{u} - \mathbf{v}$.

The vector whose entries are all zero is called the **zero vector** and is denoted by $\mathbf{0}$

Therefore, the condition

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

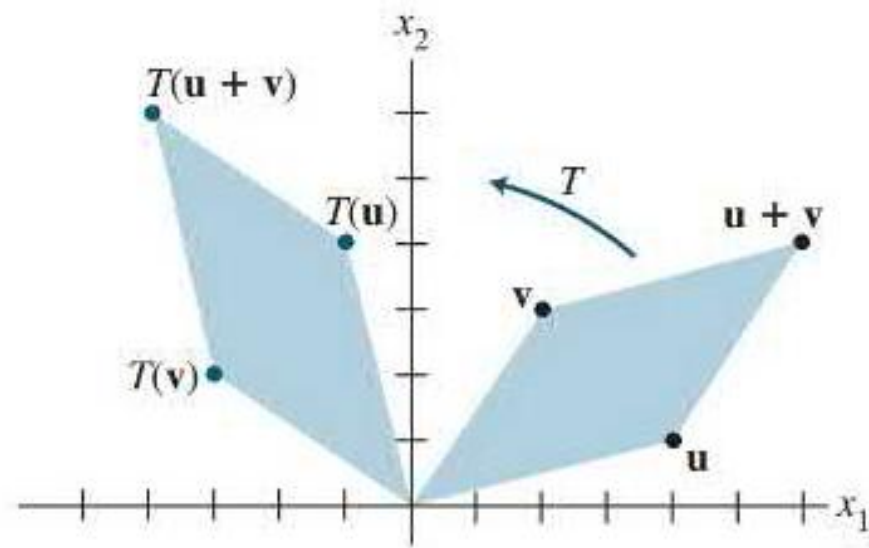
is satisfied .

Therefore, T is a linear transformation

EXAMPLE 5 Define a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.



EXAMPLE 5 Define a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

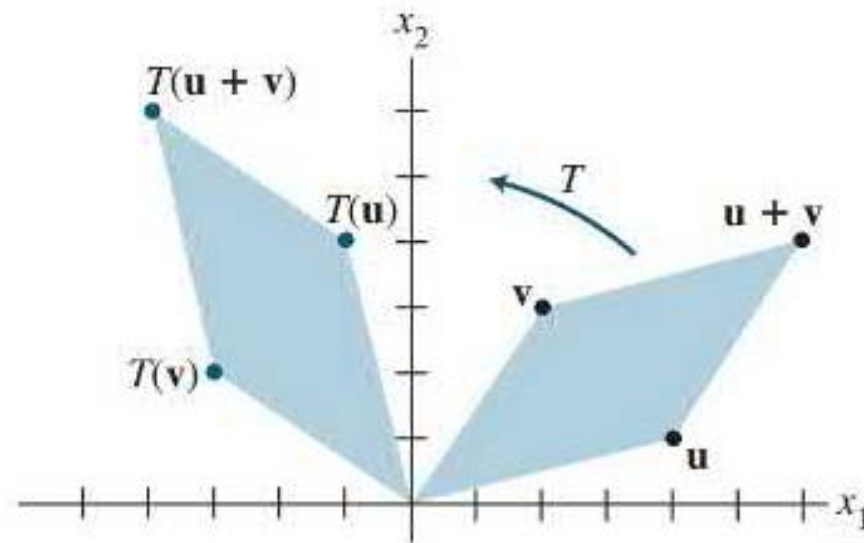
Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

SOLUTION

$$T(\mathbf{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad T(\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

$$T(\mathbf{u} + \mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

T rotates $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$ counterclockwise about the origin through 90° .



$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

In this example, the angle is 90 degrees

The Matrix of a Linear Transformation

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{bmatrix} \quad (3)$$

The Matrix of a Linear Transformation

Steps to determine the standard matrix of linear transformation A

1. Get the description of the transformation that you need to find the matrix for. That description could be in many possible ways: geometrically, formula, in words ... etc.

2. Determine the domain. If the domain is $\mathbb{R}^n$, we use the identity matrix I_n

e.g: if the domain is $\mathbb{R}^4$, we use I_4 which is this

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3. Apply the transformation (as described in Step 1) on each column of the identity matrix I_n

4. The result of transforming one column in I_n is considered a column in the matrix A

EXAMPLE 2 Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.

EXAMPLE 2 Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.

1. This transformation multiplies each vector input by 3

2. Since the domain is $\mathbb{R}^2$ (given), we use the identity matrix I_2 which is

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3. Apply the transformation on each columns of I_2 , let's call them

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

we get these results: $T(\mathbf{e}_1) = 3\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $T(\mathbf{e}_2) = 3\mathbf{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

4. Each result is a column in the matrix A :

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

the standard matrix of linear transformation

EXAMPLE 3 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , with counterclockwise rotation for a positive angle. Find the standard matrix A of this transformation.

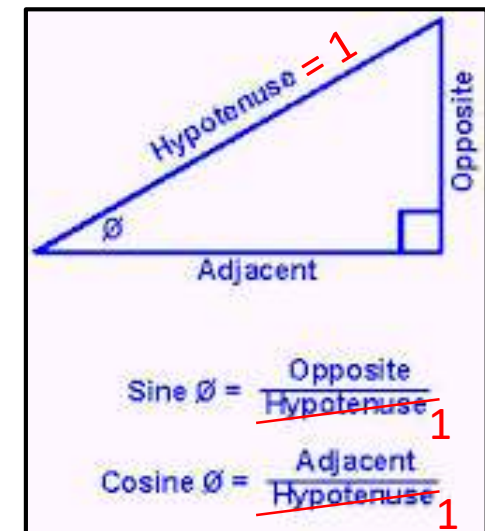
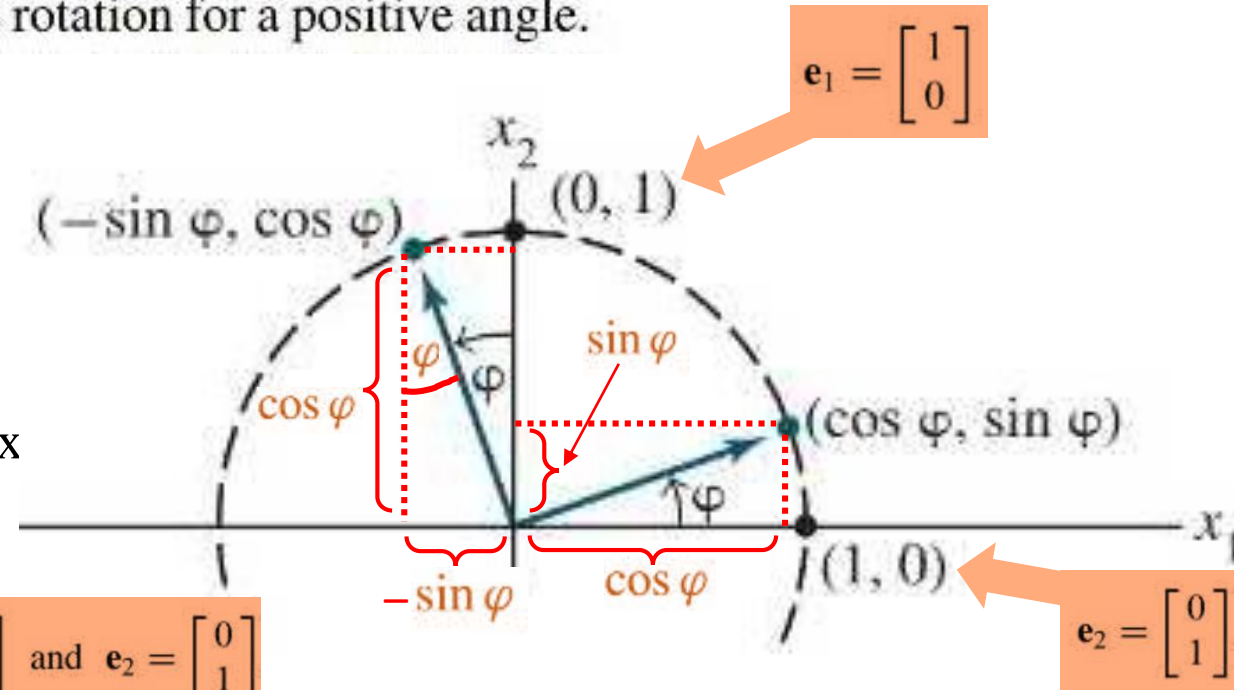
1. This transformation rotates a vector input counterclockwise. The rotation angle is any given angle φ
2. Since the domain is $\mathbb{R}^2$ (given), we use the identity matrix I_2 which is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
3. Apply the transformation on each columns of I_2
We get the results:

$$T(\mathbf{e}_1) = \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} \quad T(\mathbf{e}_2) = \begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}$$

4. Each result is a column in the matrix A :

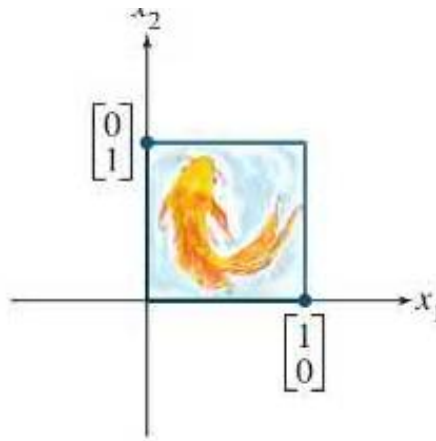
$$A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$

This is the standard transformation matrix of counterclockwise rotation in $\mathbb{R}^2$



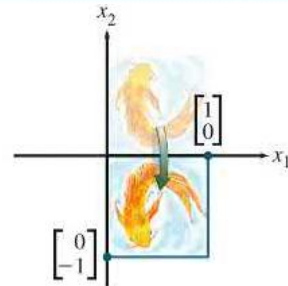
Geometric Linear Transformations of $\mathbb{R}^2$

The unit square.



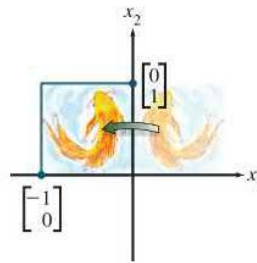
Transformation	Image of the Unit Square	Standard Matrix
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Reflection through the x_1 -axis



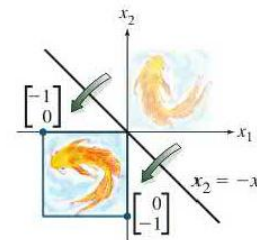
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Reflection through the x_2 -axis



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reflection through the line $x_2 = -x_1$



$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

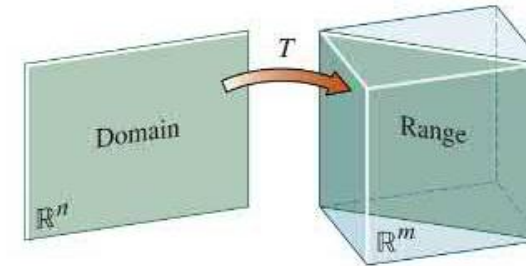
Important note:
See a list of
geometric linear
transformations
given in the handout

Existence and Uniqueness Questions

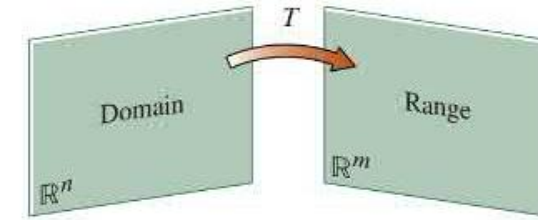
A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.

T is onto $\mathbb{R}^m$ when the range of T is all of the codomain $\mathbb{R}^m$

The question "**Does transformation T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?**" is an **existence** question $\rightarrow$ we check existence



T is not onto $\mathbb{R}^m$

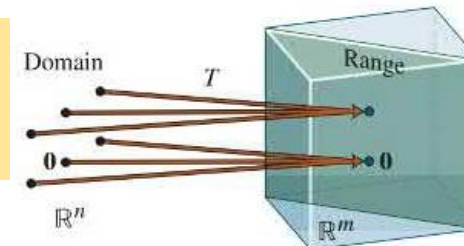


T is onto $\mathbb{R}^m$

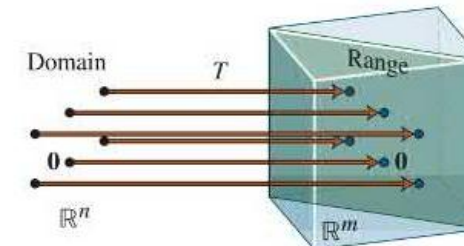
A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.

T is one-to-one if, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.

The question "**Is transformation T one-to-one?**" is a **uniqueness** question $\rightarrow$ we check uniqueness



T is not one-to-one



T is one-to-one

EXAMPLE 4 Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

$$\text{Augmented matrix} = \begin{bmatrix} 1 & -4 & 8 & 1 & b_1 \\ 0 & 2 & -1 & 3 & b_2 \\ 0 & 0 & 0 & 5 & b_3 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$?

Is T a one-to-one mapping?

EXAMPLE 4 Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

$$\text{Augmented matrix} = \begin{bmatrix} 1 & -4 & 8 & 1 & b_1 \\ 0 & 2 & -1 & 3 & b_2 \\ 0 & 0 & 0 & 5 & b_3 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$?

We check if the system $A\mathbf{x} = \mathbf{b}$ is always consistent:

We need to convert A into echelon form: It is already in echelon form

Is the **contradictory row** possible? No because A has a pivot in every **row** $\rightarrow$ **the system is consistent.**

Therefore, the transformation T maps $\mathbb{R}^4$ onto $\mathbb{R}^3$

Is T a one-to-one mapping?

We check if the system $A\mathbf{x} = \mathbf{b}$ has unique solution:

We need to convert A into echelon form: It is already in echelon form

Is the **contradictory row** possible? No, because A has a pivot in every **row** $\rightarrow$ **the system is consistent**

Are there free variables? Yes, one free variable $\rightarrow$ **the system has infinitely many solutions**

Therefore, the transformation T is not one-to-one

Existence and Uniqueness Questions

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let A be the standard matrix for T . Then:

- a. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$;
- b. T is one-to-one if and only if the columns of A are linearly independent.

EXAMPLE 5 Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show that T is a one-to-one linear transformation. Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

We don't have the standard matrix A yet. So, we find A first

Consult the following slides

The Matrix of a Linear Transformation

Steps to determine the standard matrix of linear transformation A

1. Get the description of the transformation that you need to find the matrix for. That description could be in many possible ways: geometrically, formula, in words ... etc.

2. Determine the domain. If the domain is $\mathbb{R}^n$, we use the identity matrix I_n .
e.g: if the domain is $\mathbb{R}^4$, we use I_4 which is this $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

3. Apply the transformation (as described in Step 1) on each column of the identity matrix I_n

4. The result of transforming one column in I_n is considered a column in the matrix A

Vector

If it is not the zero vector

If it is the zero vector

A Set of Two Vectors

A set of two vectors $\{v_1, v_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.

A Set of Two or More Vectors

A set of two or more vectors is linearly dependent if any of these cases is true

If and only if at least one of the vectors in the set is a linear combination of the others.

If the set contains more vectors than there are entries in each vector.

If the set contains the zero vector.

EXAMPLE 5 Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show that T is a one-to-one linear transformation. Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

We don't have the standard matrix A yet. So, we find A first

1. This transformation description: $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$

2. Since the domain is $\mathbb{R}^2$ (given), we use the identity matrix I_2

3. Apply the transformation on each columns of I_2 ,

We get the results:

$$T(\mathbf{e}_1) = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$$

$$T(\mathbf{e}_2) = \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix}$$

4. Each result is a column in the matrix A :

$$A = \begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix}$$

Q1: Is T one-to-one?

Since the columns of A are not multiple of each other, then they are linearly independent $\rightarrow T$ is one-to-one

Consult the following slides

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Q2: Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

Echelon form: Interchange R_1 and $R_3 \rightarrow R_2 = R_2 - 5R_1 \rightarrow R_3 = R_3 - 3R_1 \rightarrow R_3 = R_3 - R_2$

We don't have pivots in every row, then the **contradictory condition** is possible, then **the columns of A do not span $\mathbb{R}^3$ ($Ax=b$ is not consistent for all values of b)**, then **T does not map $\mathbb{R}^2$ onto $\mathbb{R}^3$**

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -8 \\ 0 & 0 \end{bmatrix}$$

Linear Models in Business, Science, and Engineering

Constructing a Nutritious Weight-Loss Diet

TABLE 1 The Cambridge Diet

Amounts (g) Supplied per 100 g of Ingredient				Amounts (g) Supplied by Cambridge Diet in One Day
Nutrient	Nonfat milk	Soy flour	Whey	
Protein	36	51	13	33
Carbohydrate	52	34	74	45
Fat	0	7	1.1	3

EXAMPLE 1 If possible, find some combination of nonfat milk, soy flour, and whey to provide the exact amounts of protein, carbohydrate, and fat supplied by the diet in one day (Table 1).

SOLUTION

Row reduction of the augmented matrix for the corresponding system of equations shows that

$$\begin{bmatrix} 36 & 51 & 13 & 33 \\ 52 & 34 & 74 & 45 \\ 0 & 7 & 1.1 & 3 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 0 & 0 & .277 \\ 0 & 1 & 0 & .392 \\ 0 & 0 & 1 & .233 \end{bmatrix}$$

To three significant digits, the diet requires .277 units of nonfat milk, .392 units of soy flour, and .233 units of whey in order to provide the desired amounts of protein, carbohydrate, and fat. ■

Difference Equations

If there is a matrix A such that $\mathbf{x}_1 = A\mathbf{x}_0$, $\mathbf{x}_2 = A\mathbf{x}_1$, and, in general,

$$\mathbf{x}_{k+1} = A\mathbf{x}_k \quad \text{for } k = 0, 1, 2, \dots$$

is called a **linear difference equation** (or **recurrence relation**).

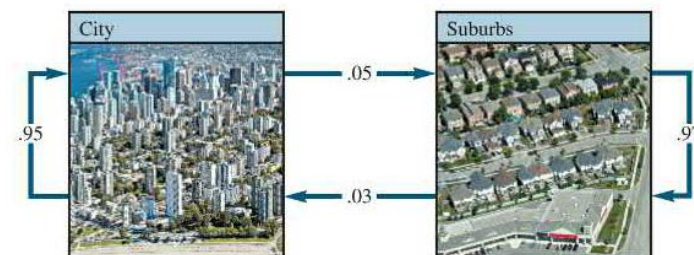
EXAMPLE 3 Compute the population of the region just described for the years 2021 and 2022, given that the population in 2020 was 600,000 in the city and 400,000 in the suburbs.

SOLUTION The initial population in 2020 is $\mathbf{x}_0 = \begin{bmatrix} 600,000 \\ 400,000 \end{bmatrix}$. For 2021,

$$\mathbf{x}_1 = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix} \begin{bmatrix} 600,000 \\ 400,000 \end{bmatrix} = \begin{bmatrix} 582,000 \\ 418,000 \end{bmatrix}$$

For 2022,

$$\mathbf{x}_2 = M\mathbf{x}_1 = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix} \begin{bmatrix} 582,000 \\ 418,000 \end{bmatrix} = \begin{bmatrix} 565,440 \\ 434,560 \end{bmatrix}$$



From:		To:
City	Suburbs	City
$\begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix}$		Suburbs